Why Our E-Learning Good Practices Work: A Quick Guide for the Team

Nov 1, 2026 · Mike MA

Every one of the 29 good practices on our e-learning design and assessment review form is backed by learning research, CAF policy, or both. This guide gives you, for each practice, what to look for, why it works in one or two plain sentences, and the source to cite when you write a review note.

Six big ideas behind the checklist

The whole form comes down to six ideas, one per domain. If you remember nothing else, remember these.

1. **Start from the job.** Training exists to close a real performance gap, for real learners in their real setting. *(Domain 1)*
2. **Line everything up with the objective.** Objective, content, practice, feedback and test should all aim at the same performance, at the same level. *(Domain 2)*
3. **Show, then let them do.** Learners need to see good performance, think and decide for themselves, and practise with support that fades over time. *(Domain 3)*
4. **Test the skill, nothing else.** A score should go up or down because of the competence, not because of reading load, tricky wording or a clumsy interface. Feedback should explain, not just mark. *(Domain 4)*
5. **Respect limited attention, and every learner.** Cut what does not help learning. Make the product accessible and of equal quality in English and French. *(Domain 5)*
6. **Check it, then keep improving it.** Experts and real learners test it before release; data and change control keep it right afterwards. *(Domain 6)*

These ideas are not just preferences. They come from CFITES doctrine, now called the CAF Individual Training and Education (IT&E) System, and from decades of learning and assessment research. The rest of this guide shows the evidence behind each practice.

How to use this guide

Each practice below has the same four parts, so you can find what you need quickly:

- **Look for:** what the practice looks like in a storyboard, module or test.
- **Why it works:** the reason, in plain language.
- **Key line:** one short quotation you can use, with its page number.
- **Cite:** the main sources, in APA 7th style.

Writing a review note. Name what you saw, say why it is good, and add the citation. For example:

> *3.5: Each module opens with three quick recall questions on earlier modules. Spaced retrieval practice is one of the best-supported learning techniques (Dunlosky et al., 2013).*

Evidence ratings. Each practice carries one or two ratings.

Rating	What it means
Strong	Many experiments and meta-analyses agree
Moderate	Sound theory with consistent, but less extensive, research support
Policy	Required or expected by CAF policy, Canadian law or a recognized standard
Practice	Established professional practice and expert consensus

A "Policy" rating is not a weak rating. It simply means the best justification is a requirement, so cite the requirement.

Want more detail? A full reference version of this guide gives the complete quotations, page numbers and discussion for every practice.

Domain 1: Performance relevance and learner context

The question: is this the right training, for these learners, in their setting?

1.1 Validated purpose and relevance

Evidence: Moderate · Policy

- **Look for:** The design names the job requirement it serves (a QS, a PO, or a needs-assessment finding). Content is framed around job tasks, not topics.
- **Why it works:** Training only fixes gaps caused by missing knowledge or skill. Starting from the job keeps every later decision tied to real performance, and lets us check later whether graduates can actually do the job.
- **Key line:** "Validation seeks to determine whether the objectives of IT&E satisfy the job-performance requirement for which the program was initially produced" (National Defence, 2009c, para. 4).
- **Cite:** Mager & Pipe (1997); Salas et al. (2012); National Defence (2025b).

1.2 Learner and context fit

Evidence: Strong · Policy

- **Look for:** A learner profile (experience, prerequisites, language, accessibility needs) and a description of the delivery setting (device, network, time available). A pre-test or skip option for experienced members.
- **Why it works:** What learners already know shapes what helps them. Step-by-step guidance helps novices but can slow down and frustrate experts.
- **Key line:** "Instructional techniques that are highly effective with inexperienced learners can lose their effectiveness and even have negative consequences when used with more experienced learners" (Kalyuga et al., 2003, p. 23).
- **Cite:** Ausubel (1968); Kalyuga et al. (2003); National Defence (2025b) for official languages.

1.3 Content necessity

Evidence: Strong

- **Look for:** Every teaching point traces to an objective. “Nice-to-know” background is cut or moved to an optional resource. Facts are backed by approved references.
- **Why it works:** Working memory is limited. Interesting but irrelevant content competes with what matters, and lowers learning.
- **Key line:** “People learn more deeply from a multimedia message when extraneous material is excluded rather than included” (Mayer, 2014, p. 63).
- **Cite:** Mayer (2014); Rey (2012); Sweller et al. (2019).

Domain 2: Outcomes, content and alignment

The question: are the objectives clear, the content right, and does everything point at the same target?

2.1 Objective quality

Evidence: Moderate · Policy

- **Look for:** Observable verbs (*identify*, *complete*, *diagnose*) instead of *know* or *understand*. Conditions (what the learner is given) and a standard (how well).
- **Why it works:** A clear objective tells the designer what to teach, the learner what is expected, and the assessor what to check. Two assessors should be able to agree on whether it was met.
- **Key line:** A PO is “a statement of the job performance standards to be met and the conditions under which they are performed” (National Defence, 2025a, para. 2).
- **Cite:** Mager (1997); Biggs (2014); National Defence (2025a).

2.2 Content accuracy and traceability

Evidence: Policy

- **Look for:** A traceability matrix (PO to EO to screen to test item). Each screen cites its reference (CFTO, doctrine, SOP) with its date. SME sign-off. The same terms in content, test and both languages.
- **Why it works:** Inaccurate content teaches errors. Traceability shows that nothing is missing, nothing is extra, and which screens to update when a reference changes.
- **Key line:** “This TP describes the Enabling Objectives and teaching points that are derived from the Performance Objectives set out in the QS” (Canadian Defence Academy, 2013, Foreword, para. 2).
- **Cite:** Canadian Defence Academy (2013); American Educational Research Association (AERA) et al. (2014), Standard 4.12.

2.3 Constructive alignment

Evidence: Strong

- **Look for:** Pick one EO and follow it: objective, demonstration, practice, feedback, test. The level should stay the same throughout, and nothing should be tested that was not taught.
- **Why it works:** If practice is easier than the test, learners are tested on something they never rehearsed. If the test is easier than the objective, a pass does not show they can do the job.
- **Key line:** "When critical features of instructional stimuli match those of assessment, effect sizes routinely reach 1.2 to 3 sigma" (Cohen, 1987, p. 16).
- **Cite:** Biggs (2014); Cohen (1987); Merrill (2002).

2.4 Instructional sufficiency

Evidence: Moderate

- **Look for:** Each EO covers the "why" (how it works), the "how" (the steps) and practice. Prerequisites come first. Attitudes are addressed through consequences and modelling, not just a policy statement.
- **Why it works:** Different kinds of learning need different conditions. Leaving out the explanation, the practice or the attitude part creates predictable gaps on the job.
- **Key line:** "Much instructional practice concentrates primarily on the demonstration phase and ignores the other phases in this cycle of learning" (Merrill, 2002, p. 44).
- **Cite:** Gagné (1985); van Merriënboer & Kirschner (2018); Merrill (2002).

Domain 3: Learning strategy and experience

The question: will the way we teach actually build the skill?

3.1 Method–outcome fit

Evidence: Strong

- **Look for:** The method matches the kind of learning. Concepts: examples and non-examples. Procedures: step-by-step demonstration, then practice. Decisions: scenarios with consequences. Behaviour: modelling.
- **Why it works:** There is no single best method. Each kind of learning needs different conditions, and novices need more guidance than experts.
- **Key line:** "Learning is promoted when the demonstration is consistent with the learning goal: (a) examples and nonexamples for concepts, (b) demonstrations for procedures, (c) visualizations for processes, and (d) modeling for behavior" (Merrill, 2002, p. 47).
- **Cite:** Gagné (1985); Merrill (2002); Kirschner et al. (2006).

3.2 Sequence and progression

Evidence: Strong

- **Look for:** Lessons open by recalling what learners already know. Key terms come before the procedure that uses them. Tasks move from simple to complex and from guided to independent.

- **Why it works:** Small steps build on what is already in memory, so learners are not overloaded. Ending with unaided performance matches what the PO requires.
- **Key line:** “Only present small amounts of new material at any time, and then assist students as they practice this material” (Rosenshine, 2012, p. 13).
- **Cite:** Merrill (2002); Rosenshine (2012); Kirschner et al. (2006).

3.3 Demonstration and modelling

Evidence: Strong

- **Look for:** Step-by-step demonstrations with the expert’s reasoning made visible. Worked examples for decisions. Where useful, a contrasting example of a common error.
- **Why it works:** For novices, studying a good example is faster and more reliable than trial and error. Showing good and poor examples together improves transfer to the job.
- **Key line:** “Learning is promoted when the instruction demonstrates what is to be learned rather than merely telling information about what is to be learned” (Merrill, 2002, p. 47).
- **Cite:** Merrill (2002); Atkinson et al. (2000); Taylor et al. (2005).

3.4 Meaningful participation

Evidence: Strong

- **Look for:** Interactions where learners decide, predict, classify or explain. Not click-to-reveal tabs that anyone could complete without thinking.
- **Why it works:** Clicking is not thinking. Learning increases as engagement moves from passive, to active, to constructive, to interactive.
- **Key line:** “The ICAP hypothesis predicts that as students become more engaged with the learning materials, from passive to active to constructive to interactive, their learning will increase” (Chi & Wylie, 2014, p. 219).
- **Cite:** Chi & Wylie (2014); Mayer (2004).

3.5 Practice and scaffolding

Evidence: Strong

- **Look for:** Recall questions on earlier content in later modules. Hints or job aids in early practice, removed later. Unaided practice at the standard before the Performance Check.
- **Why it works:** Spaced recall makes learning last. Early support makes practice possible for novices; removing it makes the final performance independent.
- **Key line:** “Testing is a powerful means of improving learning, not just assessing it” (Roediger & Karpicke, 2006, p. 249).
- **Cite:** Dunlosky et al. (2013); Roediger & Karpicke (2006); Rosenshine (2012).

3.6 Authenticity and transfer

Evidence: Strong

- **Look for:** Real CAF situations, equipment, forms and terms. Decisions members actually face. Realism spent where it matters (accurate information and consequences), not on photorealistic graphics.
- **Why it works:** People apply what they learned most readily in situations like the one where they learned it. But more realism is not automatically better.
- **Key line:** "There is a relationship between fidelity and transfer, but it is not as simple as 'more is better'" (Alexander et al., 2005, p. 6).
- **Cite:** Baldwin & Ford (1988); Grossman & Salas (2011); Alexander et al. (2005).

Domain 4: Assessment, feedback and mastery

The question: can we trust the result, and does every learner get what they need to reach the standard?

4.1 Assessment purpose

Evidence: Strong

- **Look for:** Practice checks are clearly labelled, repeatable, and give hints and explanations. The Performance Check is separate and secure, has no hints, and is scored against the PO standard.
- **Why it works:** Practice is for learning; the Performance Check supports a qualification decision. Mixing them weakens both: hints in a Performance Check inflate scores, and grading practice discourages learning from mistakes.
- **Key line:** "When the cook tastes the soup, that's formative; when the guests taste the soup, that's summative" (Stake, as cited in Scriven, 1981, p. 63).
- **Cite:** Black & Wiliam (1998); AERA et al. (2014), Standard 1.0; National Defence (2009a).

4.2 Assessment alignment and validity

Evidence: Strong

- **Look for:** A test blueprint that maps every item to an EO. Item verbs match objective verbs, and test conditions match objective conditions. If the e-learning covers only the knowledge part of a PO, the blueprint says where the performance itself is assessed.
- **Why it works:** A pass only means "qualified" if the test samples what the objective requires, at the same level. A recall quiz cannot show that someone can perform.
- **Key line:** Alignment is "the degree to which the content and cognitive demands of test questions match targeted content and cognitive demands described in the test specifications" (AERA et al., 2014, p. 216).
- **Cite:** AERA et al. (2014); Kane (2013); Krathwohl (2002).

4.3 Fairness and clarity

Evidence: Strong

- **Look for:** Clear instructions and success criteria. Short, plain question stems with no trick wording. Wrong options that are believable because they reflect real errors. No give-away clues. Interactions that are easy to operate and work with a keyboard.
- **Why it works:** If wording, reading load or the interface makes an item harder, or a clue makes it easier, the score no longer reflects the skill. Removing these irrelevant barriers is the core of test fairness.
- **Key line:** "The central idea of fairness in testing is to identify and remove construct-irrelevant barriers to maximal performance for any examinee" (AERA et al., 2014, p. 63).
- **Cite:** AERA et al. (2014), Standard 3.0; Messick (1994); Haladyna et al. (2002).

4.4 Feedback quality

Evidence: Strong

- **Look for:** Feedback for each answer option that explains why, points to the next step or review screen, and is about the task rather than the person.
- **Why it works:** "Correct" or "Incorrect" alone barely helps. Feedback that explains has the largest effect, and praise-focused feedback can even lower performance.
- **Key line:** "Feedback is more effective the more information it contains" (Wisniewski et al., 2020, p. 12).
- **Cite:** Hattie & Timperley (2007); Shute (2008); Van der Kleij et al. (2015).

4.5 Remediation and mastery

Evidence: Moderate to strong · Policy

- **Look for:** Learners who miss an objective go to targeted review, then are retested with different items on the same objective. Retest limits match the course control document.
- **Why it works:** CAF standards are fixed, while time and support can vary. Retesting with the same items measures memory of the answers, not the skill.
- **Key line:** Learners who must show mastery "should have a reasonable number of opportunities to succeed on alternate forms of the test or be provided with technically sound alternatives" (AERA et al., 2014, p. 197).
- **Cite:** Bloom (1968); Kulik et al. (1990); National Defence (2009a).

Domain 5: Multimedia, accessibility and development readiness

The question: will the build work for every learner, and can Development build it without guessing?

5.1 Multimedia purpose

Evidence: Strong

- **Look for:** Every graphic, sound and animation has a job, such as showing the equipment or the process. Narration explains the graphic instead of reading the screen aloud. Labels sit on the graphic.

- **Why it works:** Media are never neutral: each element either helps learners make sense of the content or uses up their limited attention.
- **Key line:** “People learn better from words and pictures than from words alone” (Mayer, 2020, p. 1), as long as the pictures are relevant.
- **Cite:** Mayer (2014, 2020); Rey (2012).

5.2 Cognitive manageability

Evidence: Strong

- **Look for:** Short, learner-paced segments with a “Continue” button. Key terms taught before the main procedure. Clear headings and highlighting. No auto-advance, and no duplicate text competing with narration (captions stay available as an option).
- **Why it works:** We cannot change how complex the content is, but we can change how much effort the presentation adds.
- **Key line:** “People learn more deeply when a multimedia message is presented in learner-paced segments rather than as a continuous unit” (Mayer, 2014, p. 64).
- **Cite:** Sweller et al. (2019); Mayer (2014).

5.3 Technology appropriateness

Evidence: Strong

- **Look for:** The storyboard says what each technology lets learners do (a simulation for safe practice, branching for decisions, instant scoring for feedback). A simpler option was considered. It runs on the networks and devices learners actually have.
- **Why it works:** Learning comes from the teaching method, not the delivery technology. Technology earns its cost when it makes a needed method possible.
- **Key line:** “The observed advantage for blended learning conditions is not necessarily rooted in the media used per se and may reflect differences in content, pedagogy and learning time” (Means et al., 2010, p. xv).
- **Cite:** Clark (1983); Means et al. (2010).

5.4 Accessibility and inclusion

Evidence: Policy (with UDL as the design framework)

- **Look for:** Alt text on meaningful images. Captions and transcripts. Everything works with a keyboard, in a logical order. Time limits can be extended. Clear labels on every button and field. Content offered in more than one form.
- **Why it works:** Designing for differences from the start is cheaper and better than fixing barriers later. It is also required: new federal regulations adopt the accessible ICT standard CAN/ASC-EN 301 549, which builds in WCAG 2.1 Level AA, phased in from December 2027.
- **Key line:** “Learners differ in the ways they perceive and make meaning of information” (CAST, 2024, Representation section).

- **Cite:** CAST (2024); World Wide Web Consortium (2024); Accessible Canada Act (2019); Regulations Amending the Accessible Canada Regulations (2025).

5.5 Official-language quality and equivalence

Evidence: Policy

- **Look for:** English and French developed in parallel, with the same screens, interactions, feedback and scoring. Approved bilingual terminology. Layouts that leave room for longer French text. Test items checked for equal meaning and difficulty. Both versions ready on day one.
- **Why it works:** DND's standard is "equal quality", not just "translated". A harder or weaker version in one language makes the test measure language instead of competence.
- **Key line:** "Online training that the CAF mandates or provides must always be offered in both OLs" (National Defence, 2022, para. 6.5).
- **Cite:** Official Languages Act (1985); National Defence (2022); International Test Commission (ITC, 2017).

5.6 Development clarity

Evidence: Practice

- **Look for:** Each storyboard screen names its EO and gives final text in both languages, the purpose of each media asset, every branch and its feedback, scoring and retry rules, accessibility notes and a version number.
- **Why it works:** When developers have to invent feedback, wrong options or branches, design decisions drift away from the objectives. A clear storyboard keeps the design intact through the build.
- **Key line:** "The input to the conduct phase is the training plan that describes the learning programme and the instructional materials that enable learners to achieve POs/EOs" (Department of National Defence, 2002, p. 2).
- **Cite:** Department of National Defence (2002); Tripp & Bichelmeyer (1990).

Domain 6: Quality assurance and continuous improvement

The question: has it been tested before release, and can we evaluate and improve it afterwards?

6.1 Specialist review

Evidence: Practice

- **Look for:** Documented reviews by an SME, an instructional designer, an accessibility checker, a qualified reviewer in each official language, and a technical tester on the LMS.
- **Why it works:** Each reviewer catches a different kind of problem. Expert review is the cheapest stage of testing because it happens before learners are involved.

- **Key line:** Translation and adaptation should draw on “the choice of experts with relevant expertise” (ITC, 2017, p. 11).
- **Cite:** Tessmer (1993); ITC (2017).

6.2 Learner trial and usability

Evidence: Moderate

- **Look for:** A few learners from the target group work through the module, thinking aloud, before release. Ideally one uses assistive technology. Findings are fixed and re-tested.
- **Why it works:** Designers and SMEs know the content too well to see where novices struggle. Small rounds of testing find most problems cheaply.
- **Key line:** Usability is the “extent to which a system, product or service can be used by specified users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use” (International Organization for Standardization [ISO], 2018, clause 3.1.1).
- **Cite:** Tessmer (1993); ISO (2018); Nielsen (2000).

6.3 Evaluation and transfer readiness

Evidence: Moderate

- **Look for:** Results recorded by EO. A survey that asks about job usefulness, not only satisfaction. The on-the-job behaviours each PO should produce are named, so supervisors can be asked about them later.
- **Why it works:** We can only evaluate and validate later what we plan for now. “Was it useful for my job?” predicts transfer better than “Did I enjoy it?”
- **Key line:** “Utility-type reaction measures were more strongly related to learning or on-the-job performance (transfer) than affective-type reaction measures” (Alliger et al., 1997, p. 341).
- **Cite:** Kirkpatrick Partners (n.d.); Alliger et al. (1997); National Defence (2009b, 2009c).

6.4 Revision and change control

Evidence: Practice

- **Look for:** A review log (finding, decision, change, version). Version numbers on storyboards and builds. Shared elements, such as the glossary and standard feedback, kept in one place. Every change applied to both languages and re-tested.
- **Why it works:** Without change control, fixes land unevenly: one screen is fixed and its twin is not, or the English is updated and the French is not.
- **Key line:** “The PDSA Cycle (Plan-Do-Study-Act) is a systematic process for gaining valuable learning and knowledge for the continual improvement of a product, process, or service” (The W. Edwards Deming Institute, n.d.).
- **Cite:** The W. Edwards Deming Institute (n.d.); National Defence (2025b).

6.5 Efficiency

Evidence: Moderate · Policy

- **Look for:** Only what the objectives need. A test-out option for experienced members where policy allows. The simplest technology that does the job. A product that is easy to update.
- **Why it works:** Efficiency means learning per unit of learner time and organizational effort. It does not mean "shorter is better": cutting the practice learners need is a false economy.
- **Key line:** The CAF IT&E System aims to provide training "at the right time and in the most efficient and effective manner" (National Defence, 2015, para. 4).
- **Cite:** National Defence (2015, 2025b); Paas & van Merriënboer (1993); Mayer (2014).

Suggested updates to the review form

The form is sound as it stands. These changes would make it easier to apply consistently.

Two criteria worth adding

New criterion	What it checks	Why
Scoring consistency and defensible standards (Domain 4)	Answer keys and rubrics are complete; the Performance Check pass mark is justified and documented; item pools are large enough for retests	No current criterion checks how the pass mark was set, and the testing standards require it to be documented (AERA et al., 2014, Standard 5.21)
Motivation and relevance to the learner (Domain 3)	The design gains attention, shows relevance to the learner's own job, builds confidence through achievable steps, and shows progress	Motivation affects transfer, and no current criterion addresses it (Keller, 2010)

Five quick wording tweaks

1. **3.5 Practice and scaffolding:** name both parts, spaced recall and support that fades.
2. **4.3 Fairness and clarity:** add "and free from irrelevant clues", not just irrelevant difficulty.
3. **4.4 Feedback quality:** add "explains why the response is correct or incorrect".
4. **4.5 Remediation and mastery:** add "reassessment uses different items on the same objective".
5. **5.4 Accessibility and inclusion:** add a measurable standard, "WCAG 2.1 Level AA".

Two housekeeping notes

- Criteria 4.2 and 5.5 use working titles in this guide (*Assessment alignment and validity*; *Official-language quality and equivalence*). Check them against the current form.
- Since December 2025, DAOD 5031-2 uses the name *Canadian Armed Forces Individual Training and Education System*. Consider writing "CAF IT&E System (CFITES)" on the form.

Plain-language glossary

Term	What it means
QS, TP, PO, EO, PC	Qualification Standard, Training Plan, Performance Objective, Enabling Objective, Performance Check:

Term	What it means
	the CFITES chain from job requirement to lesson to test
Alignment	Objective, teaching and test all aim at the same performance, at the same level
Validity	How far the evidence supports what we conclude from a score, for example “qualified to perform”
Construct	The competence a test is meant to measure; for a Performance Check, the PO
Construct-irrelevant difficulty or clues	Anything that makes an item harder (reading load, tricky wording, clumsy interface) or easier (give-away hints) for reasons unrelated to the skill
Formative / summative	Practice that guides the learner’s next step / a test that supports a decision
Elaborated feedback	Feedback that explains why, not just “correct” or “incorrect”
Cognitive load	Mental effort in working memory; good design cuts the effort that does not help learning
Expertise reversal	What helps a novice can slow down an expert
Scaffolding and fading	Temporary support that is gradually removed as learners improve
Spaced retrieval	Recalling earlier material at intervals, which makes learning last
Transfer	Using what was learned on the job, and keeping it up over time
Fidelity	How closely training looks, acts and feels like the real job
UDL	Universal Design for Learning: designing for learner differences from the start
WCAG 2.1 AA	The international web accessibility standard built into Canada’s accessible ICT standard

Sources (APA 7th)

Every quotation in this guide was checked against the original or a reliable reproduction on 2026-11-01. Laws and Government of Canada policies are amended from time to time; check the current version before formal use.

Accessible Canada Act, S.C. 2019, c. 10. https://laws-lois.justice.gc.ca/eng/acts/A-0.6/FullText.html

Alexander, A. L., Brunyé, T., Sidman, J., & Weil, S. A. (2005). *From gaming to training: A review of studies on fidelity, immersion, presence, and buy-in and their effects on transfer in PC-based simulations and games*. DARWARS Training Impact Group. https://www.cs.uky.edu/~sgware/reading/papers/alexander2005gaming.pdf

Alliger, G. M., Tannenbaum, S. I., Bennett, W., Jr., Traver, H., & Shotland, A. (1997). A meta-analysis of the relations among training criteria. *Personnel Psychology*, *50*(2), 341–358. https://doi.org/10.1111/j.1744-6570.1997.tb00911.x

American Educational Research Association, American Psychological Association, & National Council on Measurement in Education. (2014). *Standards for educational and psychological testing*. American Educational Research Association. https://www.testingstandards.net/uploads/7/6/6/4/76643089/standards_2014edition.pdf

Atkinson, R. K., Derry, S. J., Renkl, A., & Wortham, D. (2000). Learning from examples: Instructional principles from the worked examples research. *Review of Educational Research*, *70*(2), 181–214. https://doi.org/10.3102/00346543070002181

Ausubel, D. P. (1968). *Educational psychology: A cognitive view*. Holt, Rinehart and Winston.

Baldwin, T. T., & Ford, J. K. (1988). Transfer of training: A review and directions for future research. *Personnel Psychology*, *41*(1), 63–105. https://doi.org/10.1111/j.1744-6570.1988.tb00632.x

Biggs, J. (2014). Constructive alignment in university teaching. *HERDSA Review of Higher Education*, *1*, 5–22. https://herdsa.org.au/system/files/HERDSARHE2014v01p05_0.pdf

Black, P., & Wiliam, D. (1998). Assessment and classroom learning. *Assessment in Education: Principles, Policy & Practice*, *5*(1), 7–74. https://doi.org/10.1080/0969595980050102

Bloom, B. S. (1968). Learning for mastery. *Evaluation Comment*, *1*(2), 1–12. https://files.eric.ed.gov/fulltext/ED053419.pdf

Canadian Defence Academy. (2013). *Training plan: Military Second Language Training Program – Canadian Armed Forces French Curriculum* (A-P3-050-SLT/PH-H01). Department of National Defence. https://canadabuys.canada.ca/documents/pub/att/2017/10/12/8609a60a7ce737eeaeb5f64d195a84ee/caffc_tp_2013_final_approved_26_june.pdf

CAST. (2024). *Universal Design for Learning Guidelines version 3.0*. https://udlguidelines.cast.org

Chi, M. T. H., & Wylie, R. (2014). The ICAP framework: Linking cognitive engagement to active learning outcomes. *Educational Psychologist*, *49*(4), 219–243. https://doi.org/10.1080/00461520.2014.965823

Clark, R. E. (1983). Reconsidering research on learning from media. *Review of Educational Research*, *53*(4), 445–459. https://doi.org/10.3102/00346543053004445

Cohen, S. A. (1987). Instructional alignment: Searching for a magic bullet. *Educational Researcher*, *16*(8), 16–20. https://doi.org/10.3102/0013189X016008016

Department of National Defence. (2002). *Manual of individual training and education: Vol. 6. Conduct of instructional programmes* (A-P9-050-000/PT-006). https://canadabuys.canada.ca/sites/default/files/webform/tender_notice/76635/a-p9-050-000-pt-006-en.pdf

Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques: Promising directions from cognitive and educational psychology. *Psychological Science in the Public Interest*, *14*(1), 4–58. https://doi.org/10.1177/1529100612453266

Gagné, R. M. (1985). *The conditions of learning and theory of instruction* (4th ed.). Holt, Rinehart and Winston.

Grossman, R., & Salas, E. (2011). The transfer of training: What really matters. *International Journal of Training and Development*, *15*(2), 103–120. https://doi.org/10.1111/j.1468-2419.2011.00373.x

Haladyna, T. M., Downing, S. M., & Rodriguez, M. C. (2002). A review of multiple-choice item-writing guidelines for classroom assessment. *Applied Measurement in Education*, *15*(3), 309–333. https://doi.org/10.1207/S15324818AME1503_5

Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research*, *77*(1), 81–112. https://doi.org/10.3102/003465430298487

International Organization for Standardization. (2018). *Ergonomics of human-system interaction — Part 11: Usability: Definitions and concepts* (ISO Standard No. 9241-11:2018). https://www.iso.org/standard/63500.html

International Test Commission. (2017). *ITC guidelines for translating and adapting tests* (2nd ed., Version 2.4). https://www.intestcom.org/files/guideline_test_adaptation_2ed.pdf

Kalyuga, S., Ayres, P., Chandler, P., & Sweller, J. (2003). The expertise reversal effect. *Educational Psychologist*, *38*(1), 23–31. https://doi.org/10.1207/S15326985EP3801_4

Kane, M. T. (2013). Validating the interpretations and uses of test scores. *Journal of Educational Measurement*, *50*(1), 1–73. https://doi.org/10.1111/jedm.12000

Keller, J. M. (2010). *Motivational design for learning and performance: The ARCS model approach*. Springer. https://doi.org/10.1007/978-1-4419-1250-3

Kirkpatrick Partners. (n.d.). *The Kirkpatrick model*. Retrieved November 1, 2026, from https://www.kirkpatrickpartners.com/the-kirkpatrick-model/

Kirschner, P. A., Sweller, J., & Clark, R. E. (2006). Why minimal guidance during instruction does not work: An analysis of the failure of constructivist, discovery, problem-based, experiential, and inquiry-based teaching. *Educational Psychologist*, *41*(2), 75–86. https://doi.org/10.1207/s15326985ep4102_1

Krathwohl, D. R. (2002). A revision of Bloom's taxonomy: An overview. *Theory Into Practice*, *41*(4), 212–218. https://doi.org/10.1207/s15430421tip4104_2

Kulik, C.-L. C., Kulik, J. A., & Bangert-Drowns, R. L. (1990). Effectiveness of mastery learning programs: A meta-analysis. *Review of Educational Research*, *60*(2), 265–299. https://doi.org/10.3102/00346543060002265

Mager, R. F. (1997). *Preparing instructional objectives: A critical tool in the development of effective instruction* (3rd ed.). CEP Press.

Mager, R. F., & Pipe, P. (1997). *Analyzing performance problems, or, You really oughta wanna: How to figure out why people aren't doing what they should be, and what to do about it* (3rd ed.). CEP Press.

Mayer, R. E. (2004). Should there be a three-strikes rule against pure discovery learning? The case for guided methods of instruction. *American Psychologist*, *59*(1), 14–19. https://doi.org/10.1037/0003-066X.59.1.14

Mayer, R. E. (2014). Research-based principles for designing multimedia instruction. In V. A. Benassi, C. E. Overson, & C. M. Hakala (Eds.), *Applying science of learning in education: Infusing psychological science into the curriculum* (pp. 59–70). Society for the Teaching of Psychology. http://teachpsych.org/ebooks/asle2014/index.php

Mayer, R. E. (2020). *Multimedia learning* (3rd ed.). Cambridge University Press. https://doi.org/10.1017/9781316941355

Means, B., Toyama, Y., Murphy, R., Bakia, M., & Jones, K. (2010). *Evaluation of evidence-based practices in online learning: A meta-analysis and review of online learning studies* (Rev. ed.). U.S. Department of Education, Office of Planning, Evaluation, and Policy Development. https://www2.ed.gov/rschstat/eval/tech/evidence-based-practices/finalreport.pdf

Merrill, M. D. (2002). First principles of instruction. *Educational Technology Research and Development*, *50*(3), 43–59. https://doi.org/10.1007/BF02505024

Messick, S. (1994). *Validity of psychological assessment: Validation of inferences from persons' responses and performances as scientific inquiry into score meaning* (Research Report No. RR-94-45). Educational Testing Service. https://files.eric.ed.gov/fulltext/ED380496.pdf

National Defence. (2009a). *CATO 24-07 – Academic progress of trainees on CIC training courses*. Government of Canada. https://www.canada.ca/en/department-national-defence/services/cadets-junior-canadian-rangers/cjcr-policy/catos/volume2/24-07.html

National Defence. (2009b). *CATO 24-08 – Cadet Instructors Cadre program evaluation*. Government of Canada. https://www.canada.ca/en/department-national-defence/services/cadets-junior-canadian-rangers/cjcr-policy/catos/volume2/24-08.html

National Defence. (2009c). *CATO 24-09 – Validation of Cadet Instructors Cadre training*. Government of Canada. https://www.canada.ca/en/department-national-defence/services/cadets-junior-canadian-rangers/cjcr-policy/catos/volume2/24-09.html

National Defence. (2015). *CATO 24-02 – Cadet Instructors Cadre individual training and education*. Government of Canada. https://www.canada.ca/en/department-national-defence/services/cadets-junior-canadian-rangers/cjcr-policy/catos/volume2/24-02.html

National Defence. (2022). *DAOD 5039-6, Canadian Armed Forces training and education in both official languages*. Government of Canada. https://www.canada.ca/en/department-national-defence/corporate/policies-standards/defence-administrative-orders-directives/5000-series/5039/5039-6-canadian-armed-forces-training-and-education-in-both-official-languages.html

National Defence. (2025a). *DAOD 5031-1, Canadian Armed Forces recognition of prior learning and experience*. Government of Canada. https://www.canada.ca/en/department-national-defence/corporate/policies-standards/defence-administrative-orders-directives/5000-series/5031/5031-1-canadian-armed-forces-recognition-of-prior-learning-and-experience.html

National Defence. (2025b). *DAOD 5031-2, Canadian Armed Forces individual training and education system*. Government of Canada. https://www.canada.ca/en/department-national-defence/corporate/policies-standards/defence-administrative-orders-directives/5000-series/5031/5031-2-canadian-armed-forces-individual-training-and-education-system.html

Nielsen, J. (2000, March 18). *Why you only need to test with 5 users*. Nielsen Norman Group. https://www.nngroup.com/articles/why-you-only-need-to-test-with-5-users/

Official Languages Act, R.S.C. 1985, c. 31 (4th Supp.). https://laws-lois.justice.gc.ca/eng/acts/O-3.01/FullText.html

Paas, F. G. W. C., & van Merriënboer, J. J. G. (1993). The efficiency of instructional conditions: An approach to combine mental effort and performance measures. *Human Factors*, *35*(4), 737–743. https://doi.org/10.1177/001872089303500412

Regulations Amending the Accessible Canada Regulations, SOR/2025-255. (2025). *Canada Gazette, Part II, 159*(26). https://gazette.gc.ca/rp-pr/p2/2025/2025-12-17/html/sor-dors255-eng.html

Rey, G. D. (2012). A review of research and a meta-analysis of the seductive detail effect. *Educational Research Review*, *7*(3), 216–237. https://doi.org/10.1016/j.edurev.2012.05.003

Roediger, H. L., III, & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, *17*(3), 249–255. https://doi.org/10.1111/j.1467-9280.2006.01693.x

Rosenshine, B. (2012). Principles of instruction: Research-based strategies that all teachers should know. *American Educator*, *36*(1), 12–19, 39. https://www.aft.org/sites/default/files/Rosenshine.pdf

Salas, E., Tannenbaum, S. I., Kraiger, K., & Smith-Jentsch, K. A. (2012). The science of training and development in organizations: What matters in practice. *Psychological Science in the Public Interest*, *13*(2), 74–101. https://doi.org/10.1177/1529100612436661

Scriven, M. (1981). *Evaluation thesaurus* (3rd ed.). Edgepress.

Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research*, *78*(1), 153–189. https://doi.org/10.3102/0034654307313795

Sweller, J., van Merriënboer, J. J. G., & Paas, F. (2019). Cognitive architecture and instructional design: 20 years later. *Educational Psychology Review*, *31*(2), 261–292. https://doi.org/10.1007/s10648-019-09465-5

Taylor, P. J., Russ-Eft, D. F., & Chan, D. W. L. (2005). A meta-analytic review of behavior modeling training. *Journal of Applied Psychology*, *90*(4), 692–709. https://doi.org/10.1037/0021-9010.90.4.692

Tessmer, M. (1993). *Planning and conducting formative evaluations*. Kogan Page.

Tripp, S. D., & Bichelmeyer, B. (1990). Rapid prototyping: An alternative instructional design strategy. *Educational Technology Research and Development*, *38*(1), 31–44. https://doi.org/10.1007/BF02298246

Van der Kleij, F. M., Feskens, R. C. W., & Eggen, T. J. H. M. (2015). Effects of feedback in a computer-based learning environment on students' learning outcomes: A meta-analysis. *Review of Educational Research*, *85*(4), 475–511. https://doi.org/10.3102/0034654314564881

van Merriënboer, J. J. G., & Kirschner, P. A. (2018). *Ten steps to complex learning: A systematic approach to four-component instructional design* (3rd ed.). Routledge. https://doi.org/10.4324/9781315113210

The W. Edwards Deming Institute. (n.d.). *PDSA cycle*. Retrieved November 1, 2026, from https://deming.org/explore/pdsa/

Wisniewski, B., Zierer, K., & Hattie, J. (2020). The power of feedback revisited: A meta-analysis of educational feedback research. *Frontiers in Psychology*, *10*, Article 3087. https://doi.org/10.3389/fpsyg.2019.03087

World Wide Web Consortium. (2024). *Web Content Accessibility Guidelines (WCAG) 2.2* (A. Campbell, C. Adams, R. Bradley Montgomery, M. Cooper, & A. Kirkpatrick, Eds.). https://www.w3.org/TR/WCAG22/